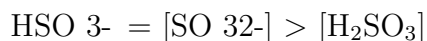
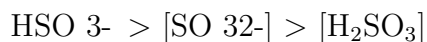


Multiple Choice (10 marks)

1. Calculate the normality of a solution containing 2.45 g of sulfuric acid in 2.00 liters of solution. (MW of $\text{H}_2\text{SO}_4 = 98.1$.)
 - (a) 0.0125 equiv/liter
 - (b) 0.060 equiv/liter
 - (c) 0.050 equiv/liter
 - (d) 0.100 equiv/liter
 - (e) 0.125 equiv/liter
2. Calculate the equilibrium polarization of ^{13}C nuclei in a 20.0 T magnetic field at 300 K.
 - (a) 7.39×10^{-5}
 - (a) 9.12×10^{-5}
 - (a) 3.43×10^{-5}
 - (a) 6.28×10^{-5}
 - (a) 1.71×10^{-5}
3. If we locate an electron to within 20pm, then what is the uncertainty in its speed? \$
 - (a) $8.110^6 \text{ m} \cdot \text{s}^{-1}$
 - (b) $9.410^7 \text{ m} \cdot \text{s}^{-1}$
 - (c) $6.210^7 \text{ m} \cdot \text{s}^{-1}$
 - (d) $1.110^8 \text{ m} \cdot \text{s}^{-1}$
 - (e) $3.710^7 \text{ m} \cdot \text{s}^{-1}$
4. Compare the bond order of He_2 and He_2^+ .
 5. Bond order of He_2 is 0 and He_2^+ is 0.5
 5. Bond order of He_2 is 0.5 and He_2^+ is 0
 5. Bond order of He_2 and He_2^+ is 0.5
 5. Bond order of He_2 is 2 and He_2^+ is 1.5
 5. Bond order of He_2 is 1 and He_2^+ is 0

A solution of sulfurous acid, H_2SO_3 , is present in an aqueous solution. Which of the following represents the concentrations of three different ions in solution?



True / False (10 marks)

Answer True or False

6. True or False: The volume of a cube with edges measuring 150.0 mm is 450 cubic centimeters.
7. True or False: The molecule NO_2 exhibits sp^2 hybridization around its central nitrogen atom.
8. True or False: The recoil energy of the atom is 400 eV, which is the expected value for such emissions in similar scenarios.
9. True or False: The reduction of $\text{NO}(\text{g})$ typically results in the formation of nitrogen dioxide $\text{NO}_2(\text{g})$ as a product.
10. True or False: Electrons can occupy the same orbital if they possess identical quantum numbers.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between temperature and the standard molar enthalpy of sulfur dioxide gas, calculating the increase in enthalpy as the temperature rises from 298.15 K to 1500 K.
12. Discuss the conditions under which precipitates of ZnS and CuS would form when H_2S is bubbled into a solution containing Zn^{2+} and Cu^{2+} ions at 0.02 M each, and analyze the implications of their solubility products.

End of Paper